MS101: Makerspace Laboratory

Lecture 6: Manufacturing Techniques

Ref:
[1] Groover, Fundamentals of Modern Manufacturing
[2] Kalpakijan and Schmidt, Manufacturing Technology
[3] Websites and Videos available on the Internet

Acknowledgement: Prof. S. P. Karagadde
for contents in some of the slides

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Criteria In Manufacturing a Product

- Economy
- Quality
- Mass Scale
- Fulfil the objective
- Durable
- Maintenance / Service
- Is the product exactly what you had desired / dreamt of ?
- Is it better/Is it just right/Is it worse than expected?
- Is the product good for its entire lifetime ? (During purchase, During use, storage, Maintainability, Serviceability, durability, efficiency of operation etc.)
- What is the resale value?

Structural or load bearing component

https://grabcad.com/library/connection-rod-crank-shaft-and-piston-assembly-1

How can this be manufactured? – if so, is a million parts possible per year?

Feasibility and Scale

Contents

- Introduction to manufacturing (15 min)
  - Alternate processing routes
  - Net Shape processes vs. Non-Net shape processes
  - Top Down Approaches :
    - CNC machining
  - Bottom-up approaches
    - Casting
    - Forming
    - 3D Printing
- Laser cutting, engraving (vector cut, rastering) (15 mins)
- Additive manufacturing (10 mins)
- Joining Methods (20 min)
  - Soldering (electronic components and metallic materials) and welding
  - Mechanical fastening (Bolts, Rivets, Dowels,Screw types)
  - Solid state joining including adhesive bonding, rivetting

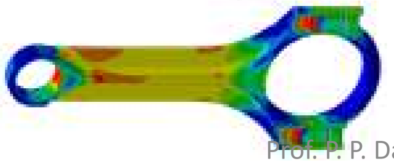
Context:
Engineering a structural component

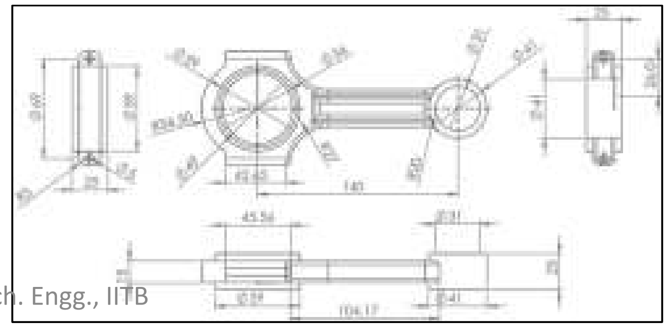
Idea
- Converting linear motion to rotation
- Sketches and Drawing

Analysis of strength
- Check whether a material can satisfy operational requirement
- Conditions of high temperature etc

Sketches and Drawing
- Geometric compliance with the requirement
- 3D Views for manufacturing

Manufacture
- Choose a method that gives the required properties
- Need to have some idea as to what can be used?





https://grabcad.com/library/connection-rod-crank-shaft-and-piston-assembly-1

From Raw Material to the final Product

- DIFFERENT FORMS OF RAW MATERIAL

- Block of raw material
- Plate
- Sheet
- Rod
- Hollow Tube
- Particles of metallic powder

FINAL GOAL : STRUCTURAL STRENGTH

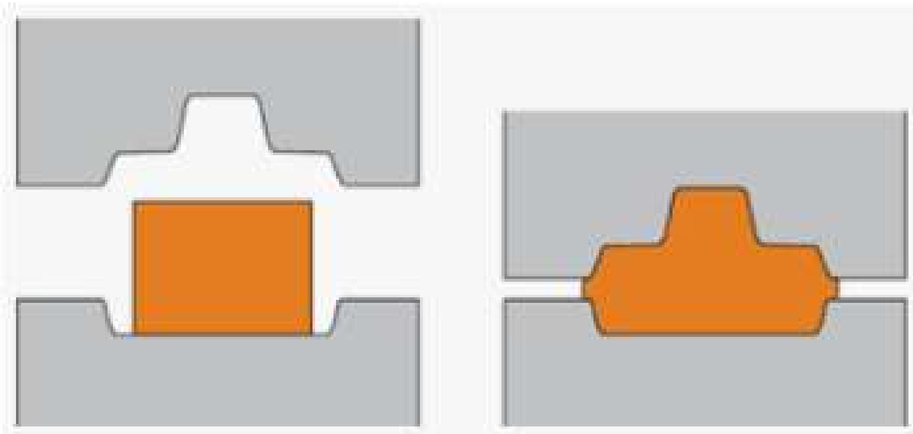
Introduction: Manufacturing

- Manufacturing is physical realization.
- We will discuss here only mechanical/discrete/solid manufacturing It is shape realization. That means our focus is limited to solids.
- Our aim is to achieve the geometry (external – shape/form, size, color) & matrix (internal – composition, porosity etc.) of the solid.
- The following two most common 'mechanisms' associated with manufacturing

1. Heat
2. Force or pressure

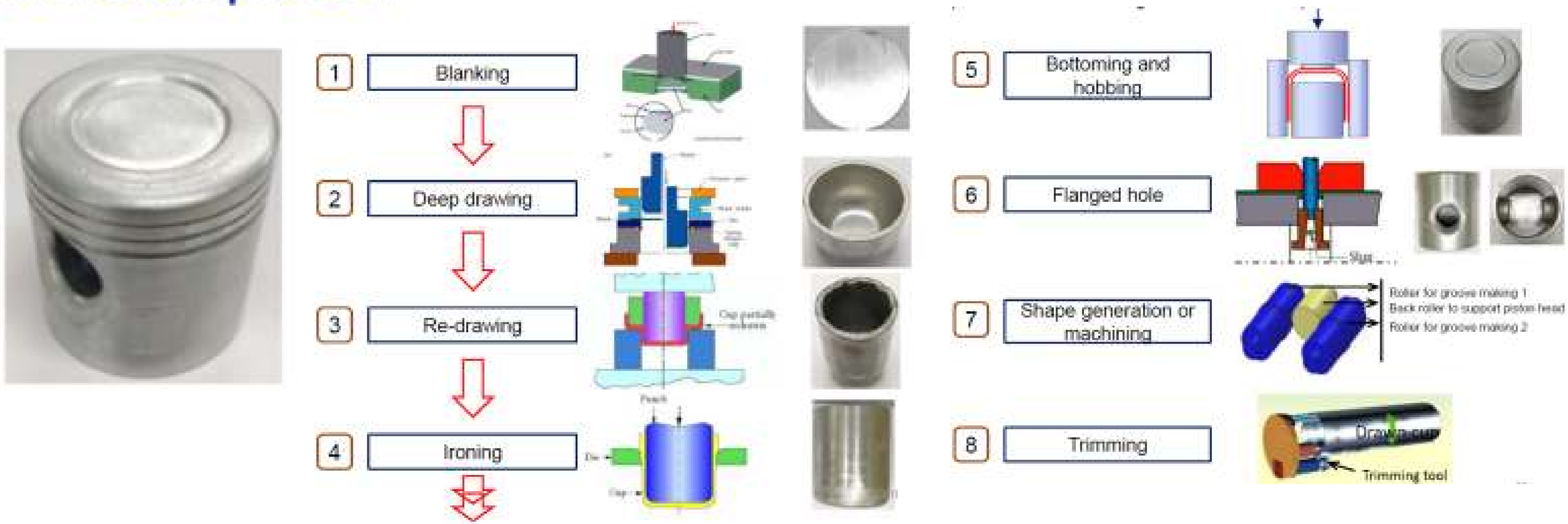
Physical considerations

- Stress vs deformation behavior
- Elastic, plastic and Viscoplastic deformation
- Microstructural effects – recrystallization
- Strengthening mechanisms

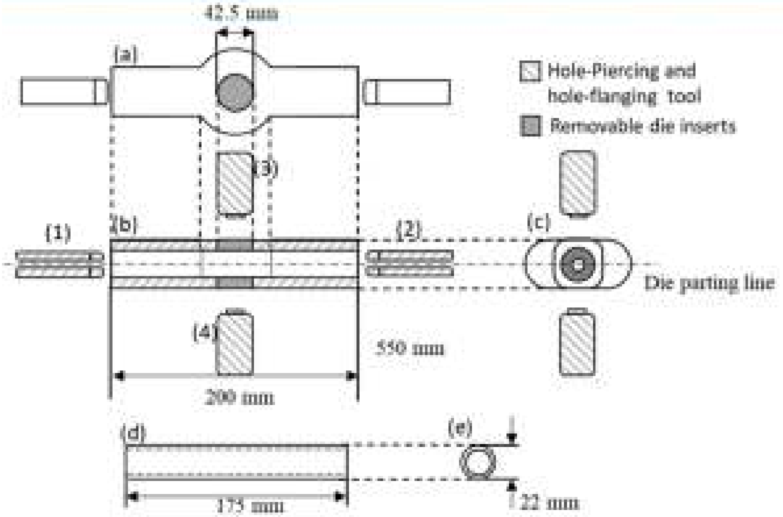


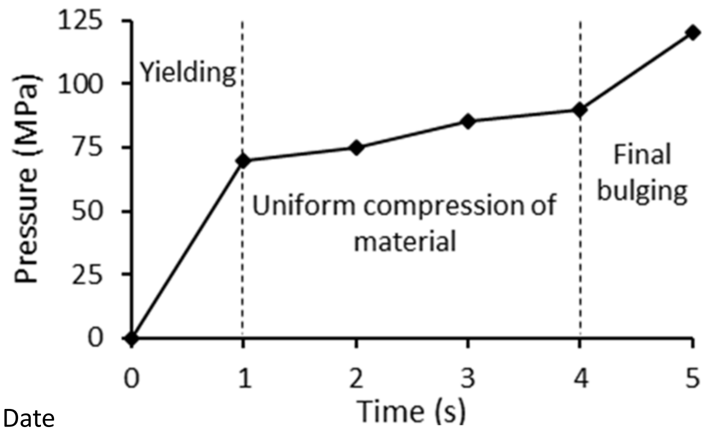
Other possibilities ?

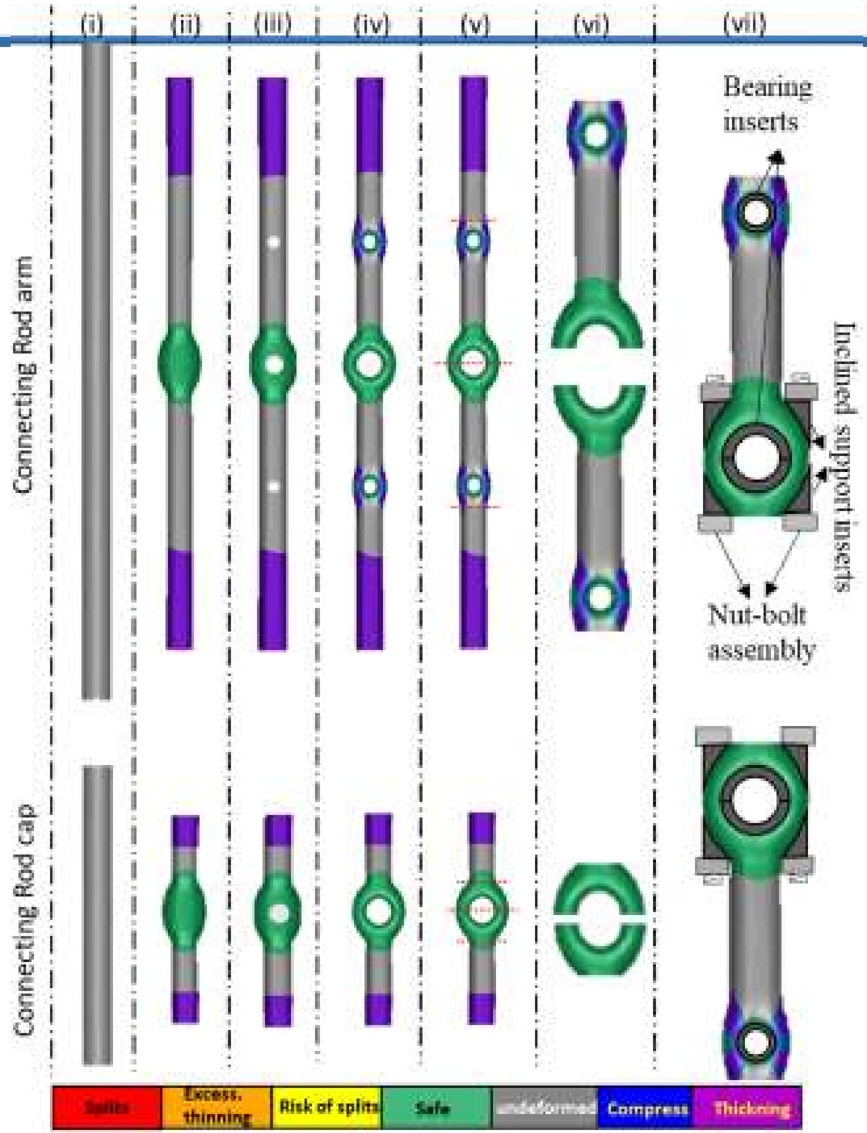
Sheet metal piston



Light Weight Connecting Rod

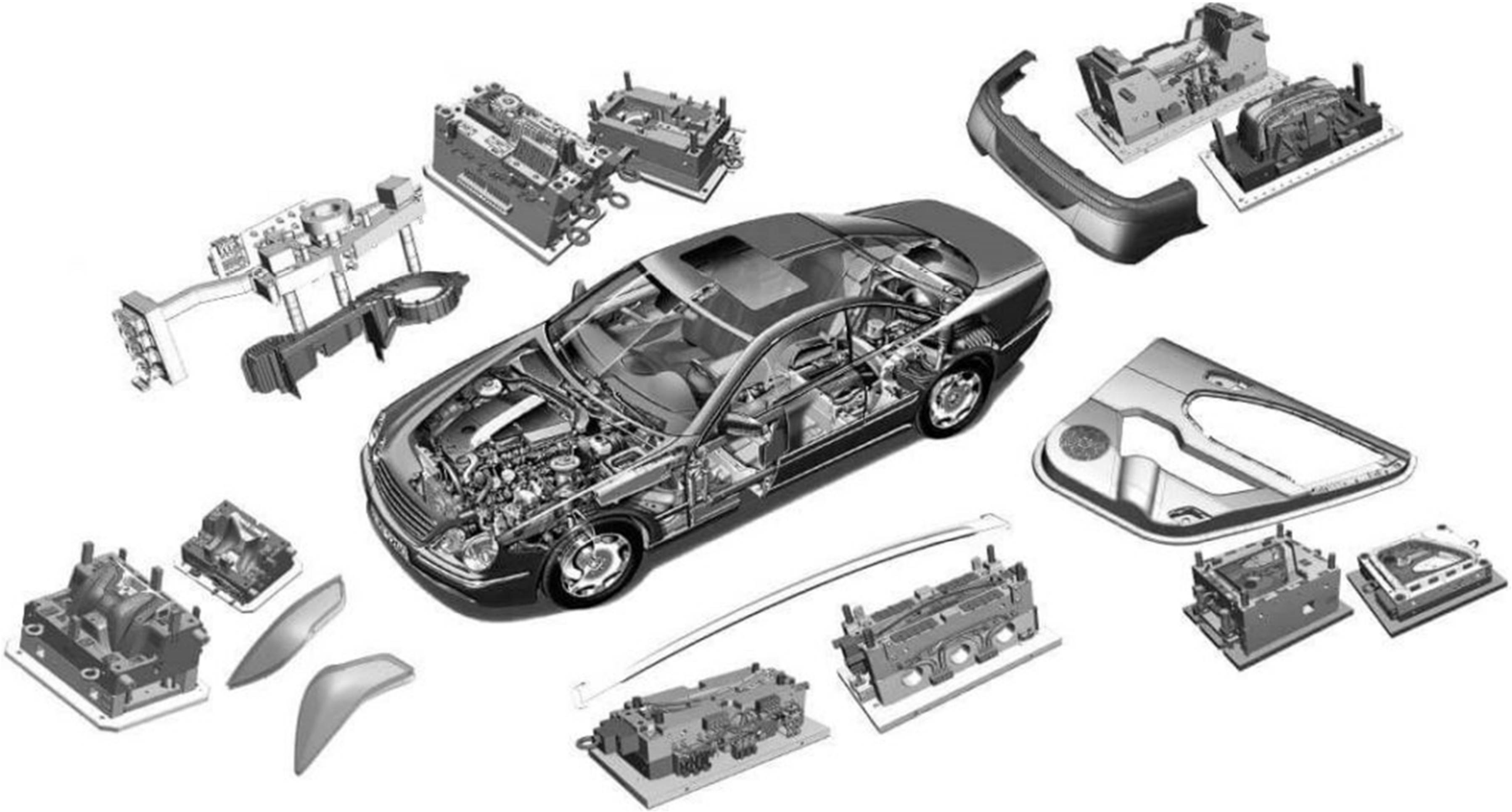






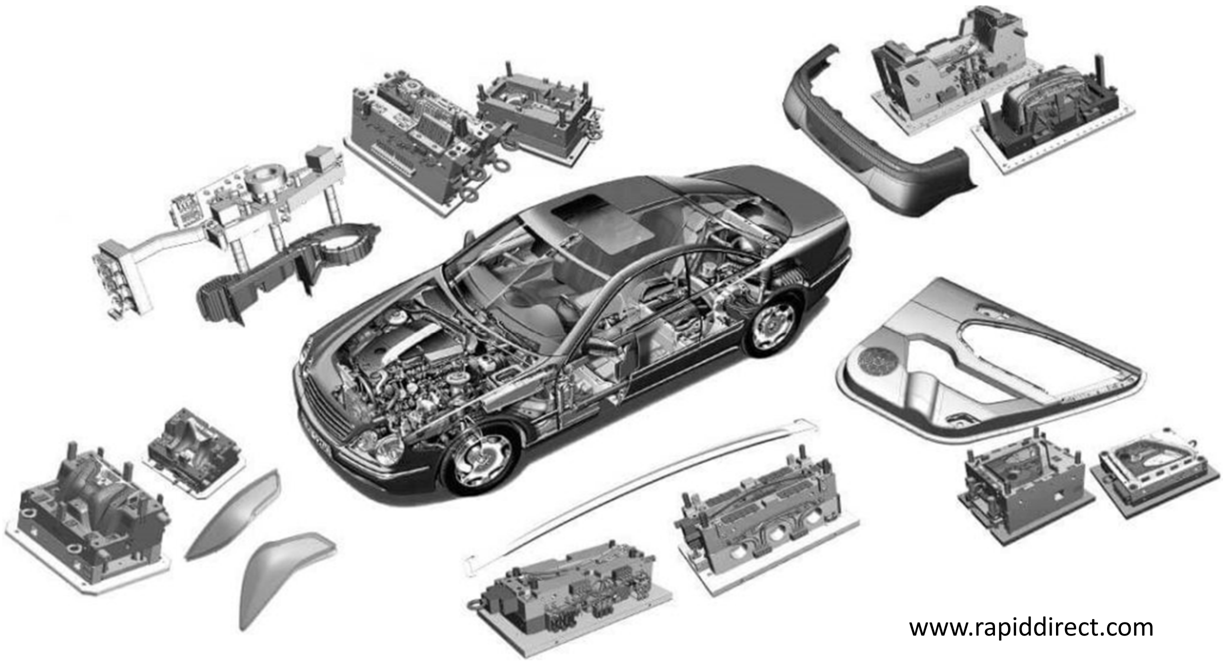
Ankit Pandey and P. P. Date

Which materials? Which Processes?

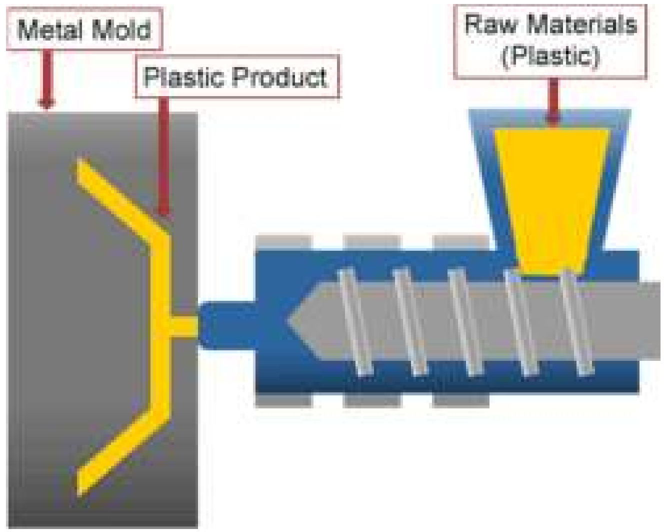


Injection Moulding of Polymeric materials

- Inject molten plastic materials into a mold cavity
- The melted plastic then cools and hardens, and the manufacturers extract the finished part.



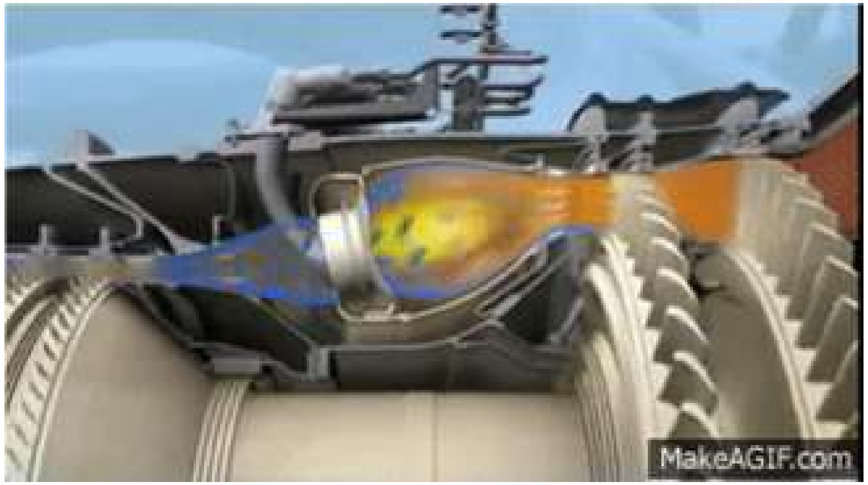
www.rapiddirect.com



Based on the starting material

- **Liquid** – pour the liquid metal into the shape and allow it to freeze (casting)
  - Requires heating alone
- **Semi-solid and 'softened' solid** – Deform the material 'forcefully' to realize the shape (forming and forging)
  - Heating and force/pressure
- **Solid:**
  - Powder/wire: heat and pressurize the powders or melt the powder/solid to deposit (**Sintering, 3D printing**)
  - Blocks: Apply force to change the shape of the solid (**Deformation**) OR Remove the unwanted portion by cutting (**Machining**) or Both in a sequence
- **Joining / Assembly:** Fastening, Fusion joining - welding

Another one

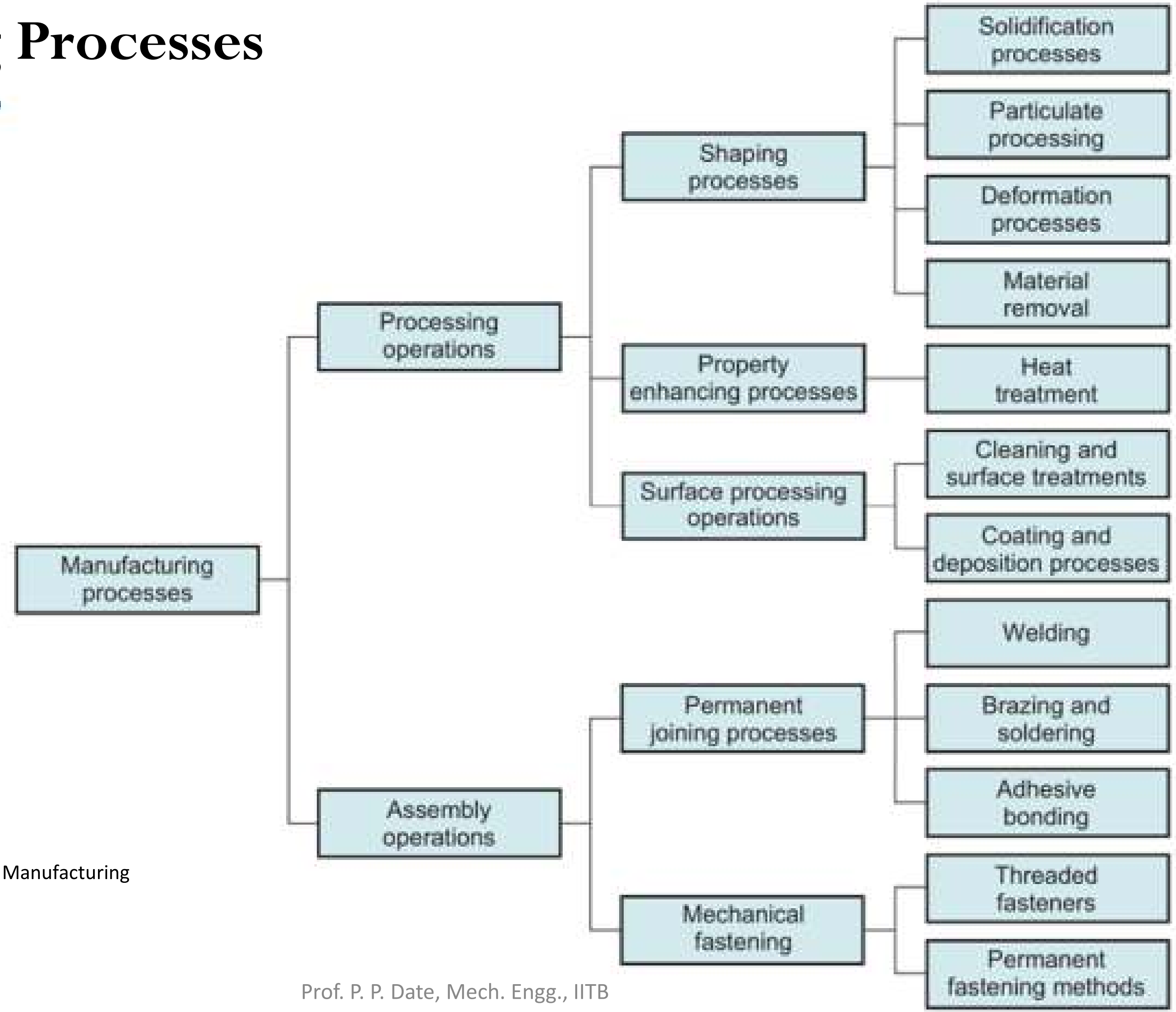


Turbine blade – external and internal

Turbojet engine – loading on blades

How can this be manufactured? – if so, ten thousand parts possible per year?
Feasibility and Scale

Manufacturing Processes



Source: M Groover: Fundamentals of Modern Manufacturing

Typical Process

- Cast by pouring liquid into shapes
- Form by compressing softened material into shapes
- 3D-Printing: Form shapes directly by adding layer by layer

Casting

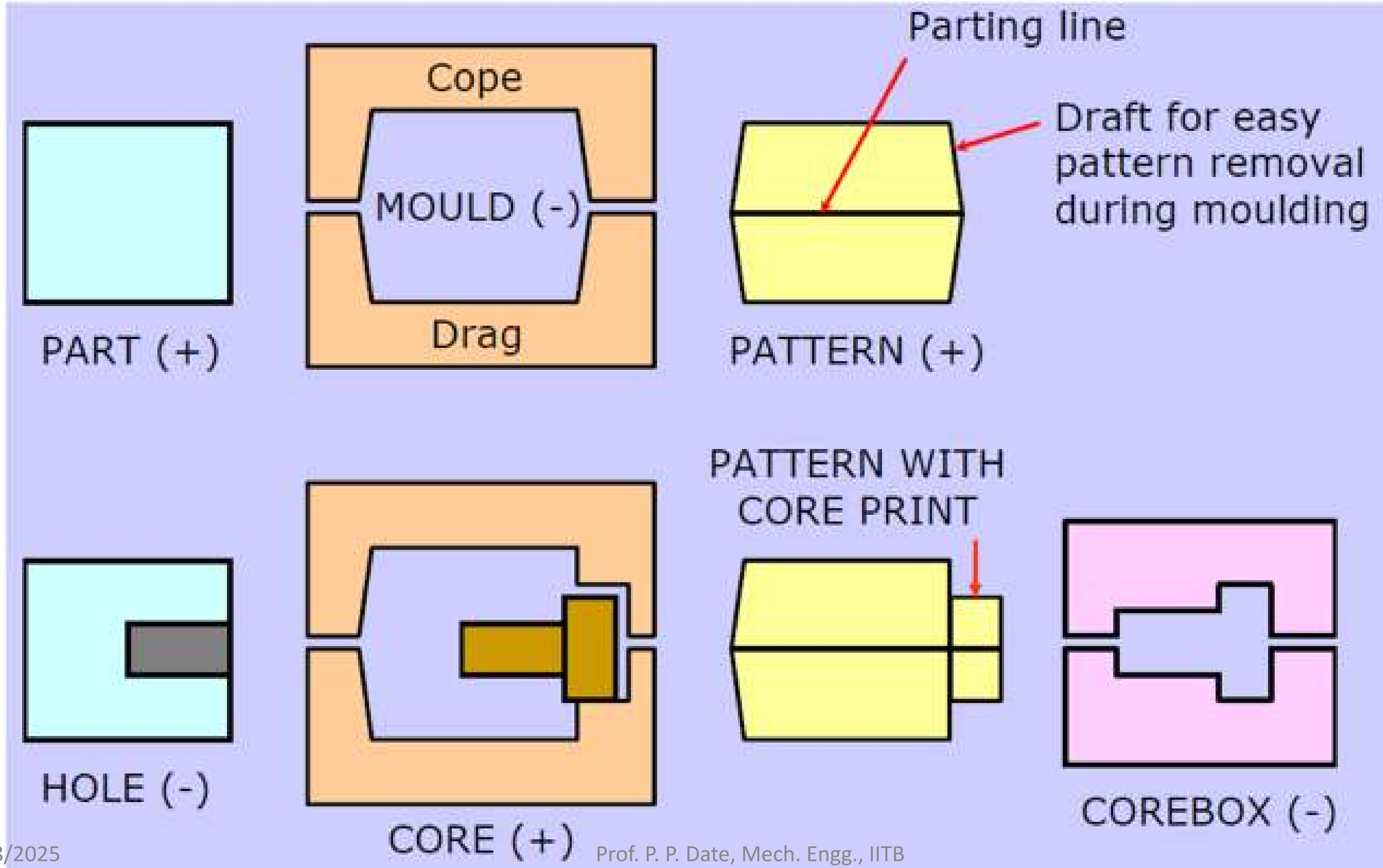
- Casting is the production of a solid metal item by allowing liquid metal to solidify in a shaped mould/die

- “You are never more than 10 feet away from a casting”

- Cars, dishwashers, fire hydrants, storm sewer inlets, grills, golf clubs, tools, kitchenware, or wheels.




Sand casting – Simplest and DIY-type



Laser Cutting

- CO_2 Lasers: The laser is generated from electrically stimulated gas mixtures (mostly comprising of carbon dioxide). CO_2 lasers are the most common types of laser cutters because they are low power, relatively inexpensive, efficient, and can both cut through and raster a wide variety of materials.

- Materials: wood, paper based products (cardboard, etc), leather, acrylic, glass, some plastics, and some foams (can raster on anodized metals)

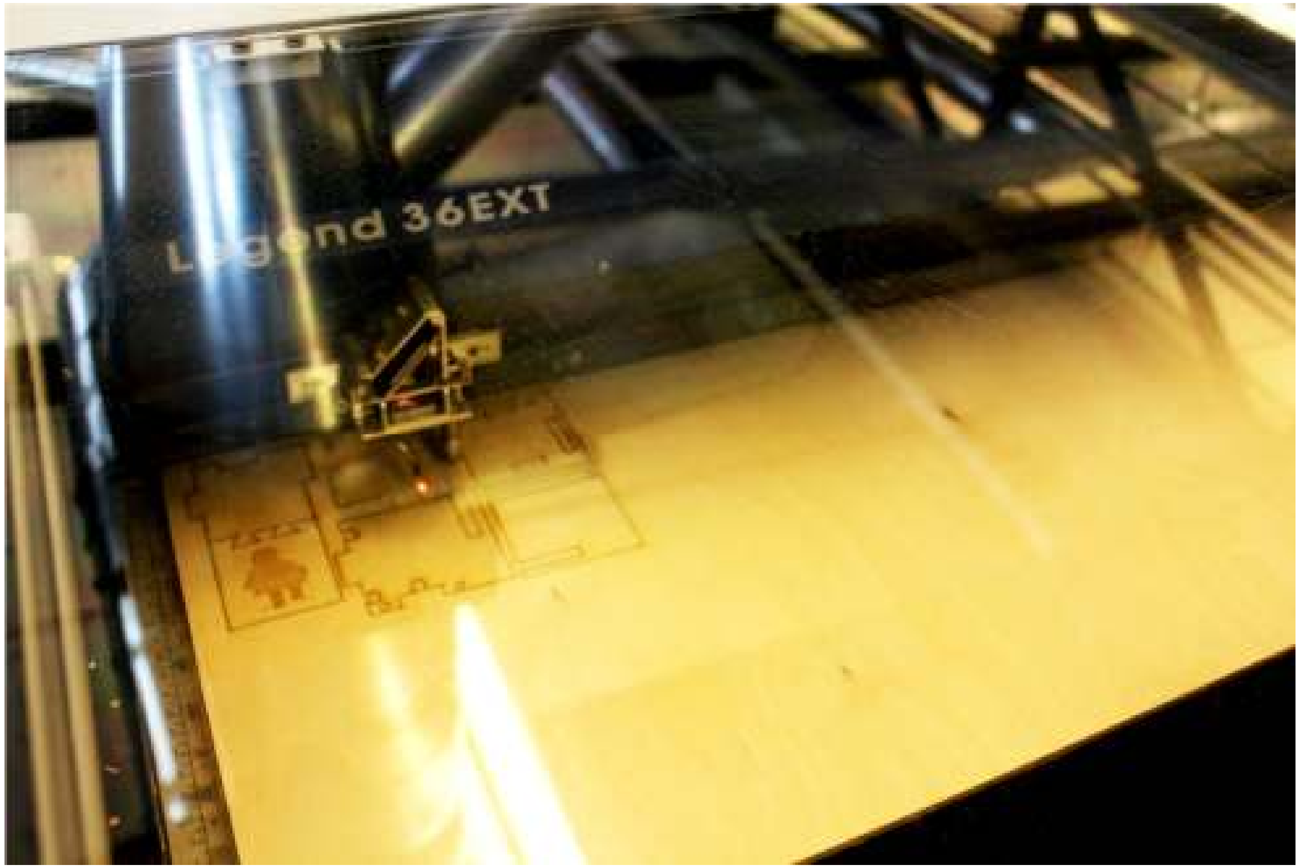
- Neodymium Lasers: The laser is formed from neodymium doped crystals. These lasers have a much smaller wavelength than CO_2 lasers, meaning they have a much higher intensity, and can thus cut through much thicker, stronger materials. However, because they are so high power, parts of the machine wear and tend to need replacing.

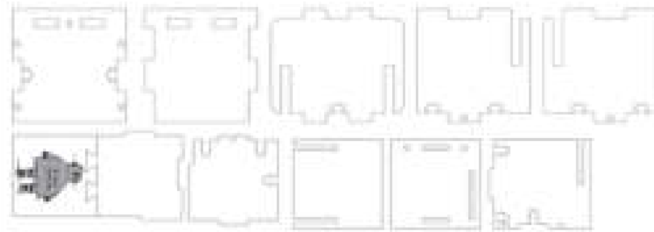
- Materials: metals, plastics, and some ceramics

- Fiber Lasers: These lasers are made from a "seed laser", and then amplified via special glass fibers. The lasers have an intensity and wavelength similar to that of the neodymium lasers, but because of the way they are built, they require less maintenance. These are mostly used for laser marking processes.

- Materials: metals and plastics

More on Laser Cutting





During a cutting operation, the cutting head fires a continuous laser at the material to slice through it. In order to know where to cut, the laser cutter driver reads all of the vector paths in the designed piece. Once you send your file to a laser cutter, only lines that register as only hairline or vector graphics with the smallest possible line thickness will be cut by the laser. Vector cutting is normally used for cutting out the outline of the part as well as any features or holes that you want to cut out of the material.

Laser Rastering

Instead of cutting all the way through the workpiece, the laser will burn off the top layer of the material you are cutting to create two color (and sometimes grayscale) images using the raster effect. In order to raster materials, the laser will usually be set to a lower power than it would when vector cutting material, and instead of shooting down a pulsing beam, it creates fine dots at a selected DPI (dots per inch) so that the laser doesn't really cut all the way through.

Power: How strongly the laser fires. A high power will cut through stronger, thicker material, but may end up burning thinner, more flammable stock. During rastering, higher power will burn more layers off of the material, creating a darker image.

Speed: How fast the head of the laser cutter moves along its gantry. A high speed will cut faster, but may not cut all the way through if you have thicker or stronger materials. A low speed has the potential to burn or melt the edges of the material. During raster operations, the laser moves back and forth very fast, so a high speed on a large piece may wear out the gantry.

Frequency (only for cutting): Determines how fast the laser pulses during a cutting operation. The laser turns on and off rapidly when it makes cuts, so a higher frequency will create a cleaner cut, but if the material is flammable it may end up catching fire, so a lower frequency would be preferable.

Additive Manufacturing (AM) / Rapid Prototyping (RP) / 3D Printing (3DP)

- Additive Manufacturing (AM) refers to a production process in which components are created layer by layer on the basis of digital 3D design data

Additive Manufacturing / 3D Printing

The Economist | Menu | Weekly edition | Search

Schumpeter | Additive manufacturing

Print me a jet engine

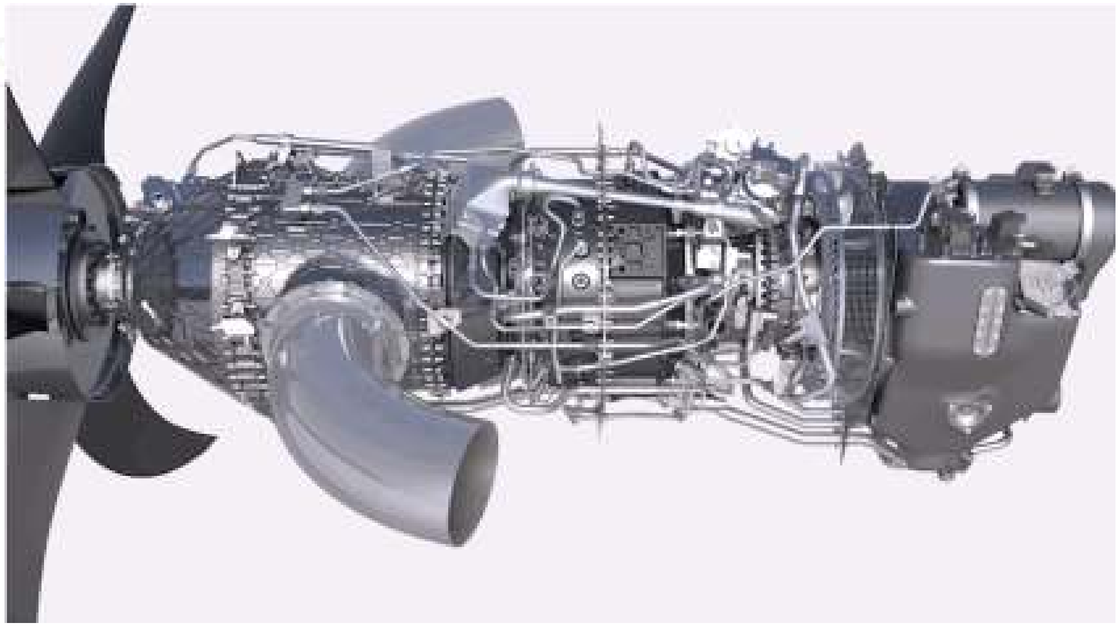
The acquisition of Morris Technologies is further proof that product innovation will increasingly go hand-in-hand with manufacturing innovation

Nov 22nd 2012

By P.M.

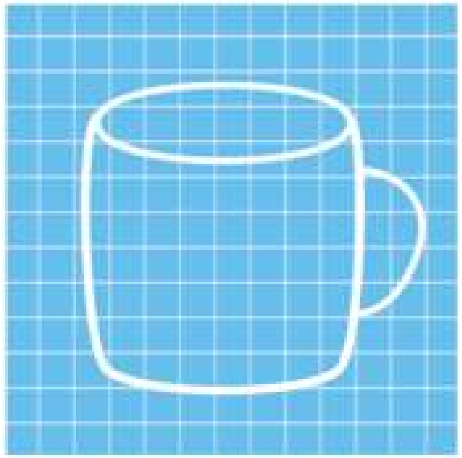
Turboprop Engine: GE Aviation

3D printing enabled the team to **combine 855 separate components into just 12**



3DP: Basic Process Flow

- Modelling in CAD
- Generating an STL or 3MF file
  - Surface geometry of 3D object
  - Orientation and Support structure
- Slicing (machine specific instructions)
  - Transforming an STL file into G-code (printer motions)
- Printing
- Post-processing
  - Finishing
  - Support structure removal
  - Heat treatment

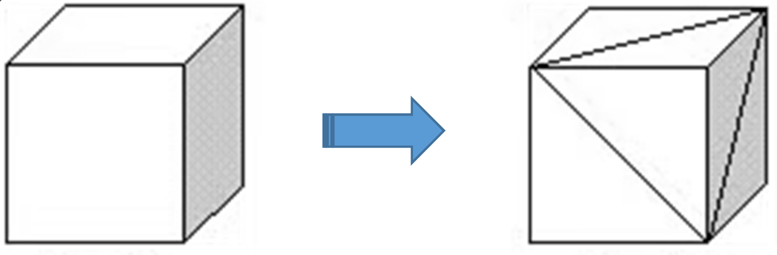


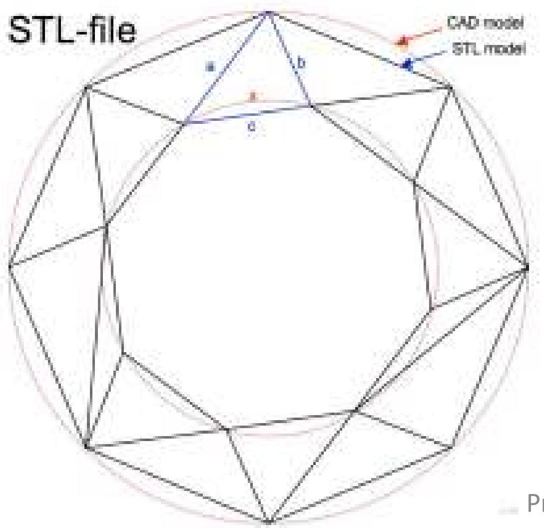
Start with a 3D CAD file either by creating the 3D model or scanned with a 3D scanner

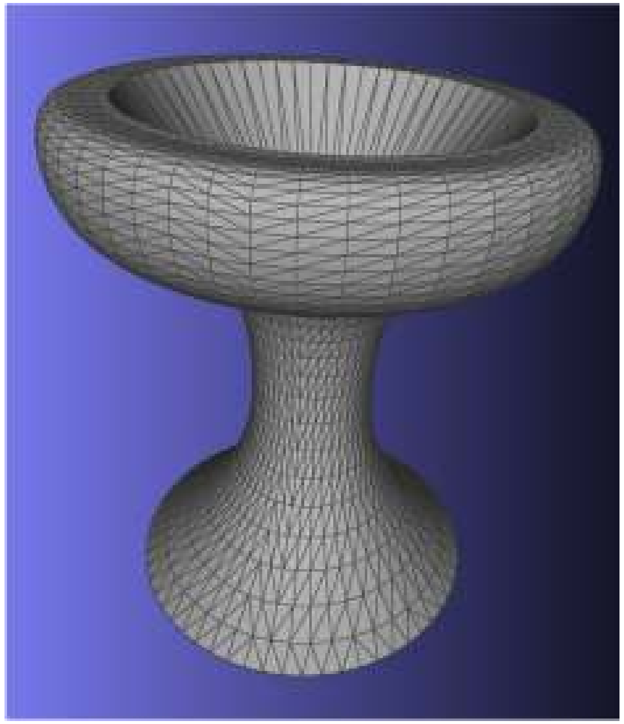
Source: https://3dprintingindustry.com

.STL files

- STereo Lithography or Standard Tessellation Language
- Approximates a 3D model by its outer surfaces using multiple triangles





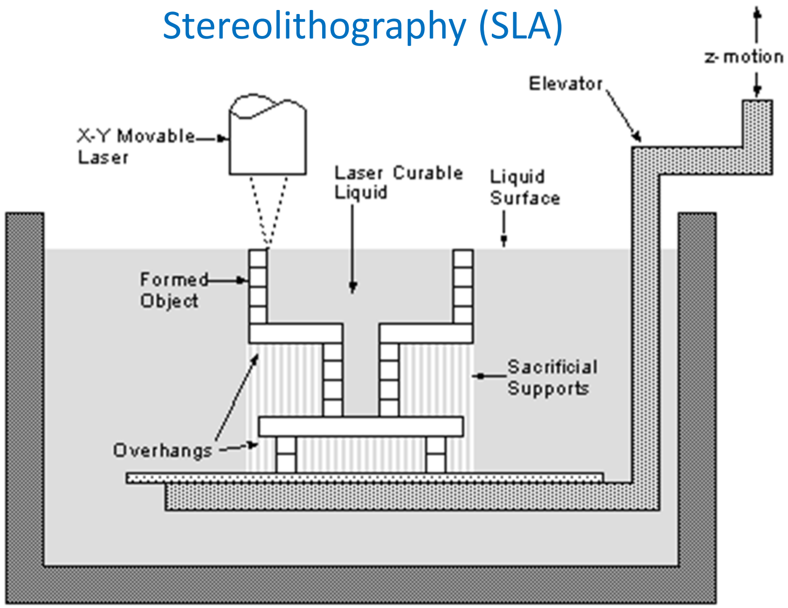


Source: Wikimedia Commons

3D Printing: Approaches

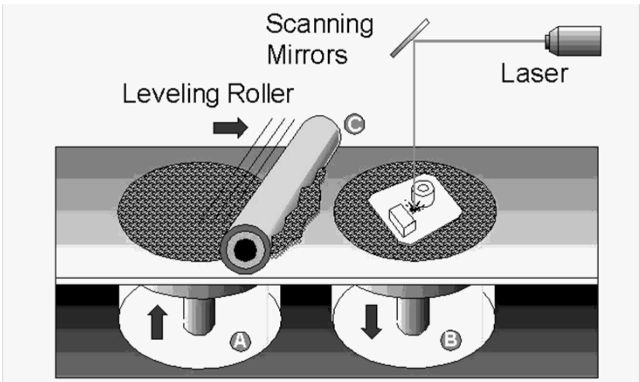
- Photopolymerization: Stereolithography (SLA)
- Powder fusion: Selective Laser Sintering (SLS)
- Fused polymer extrusion: Fused Deposition Modelling (FDM)

Stereolithography (SLA)



https://youtu.be/8a2xNaAkvLo

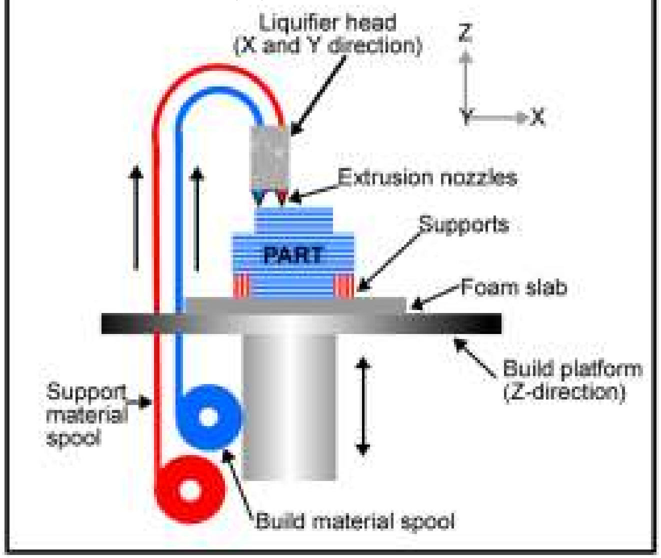
Selective Laser Sintering (SLS)



https://youtu.be/XyFSolk5OW8

Fused Deposition Modelling (FDM)

Source: Materials 2014, 7(12), 8168-8188



FDM

https://youtu.be/WHO6G67GJbM

AM examples

3D Printing of plastics - FDM

https://www.youtube.com/shorts/YyWqD6Zcqjw

https://www.youtube.com/shorts/WUCBeeATyMU

3D Printing of metals - Directed energy deposition of powders and wires

https://www.youtube.com/watch?v=oL7bMhPTtDI

Assembly/Joining

Joining Methods

Adhesive Bonding

Welding

Mechanical Fastening

Nails, rivets, bolts, etc.

Fusion

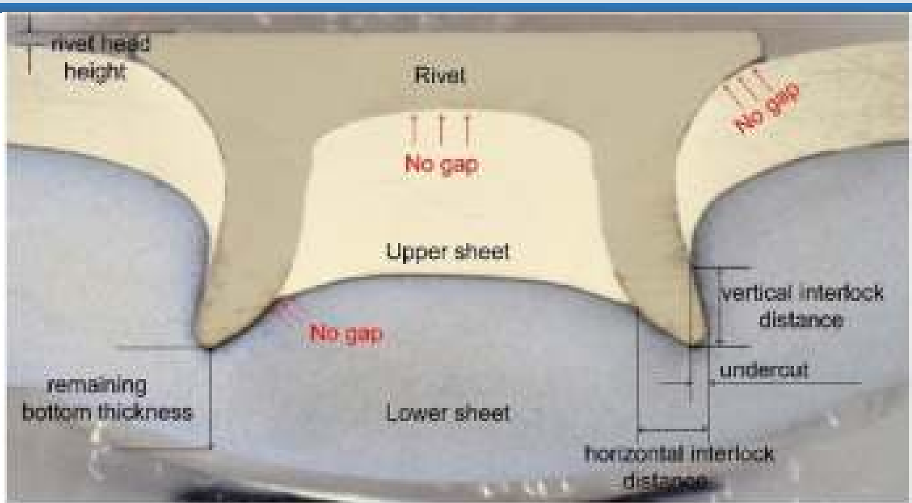
Brazing & Soldering

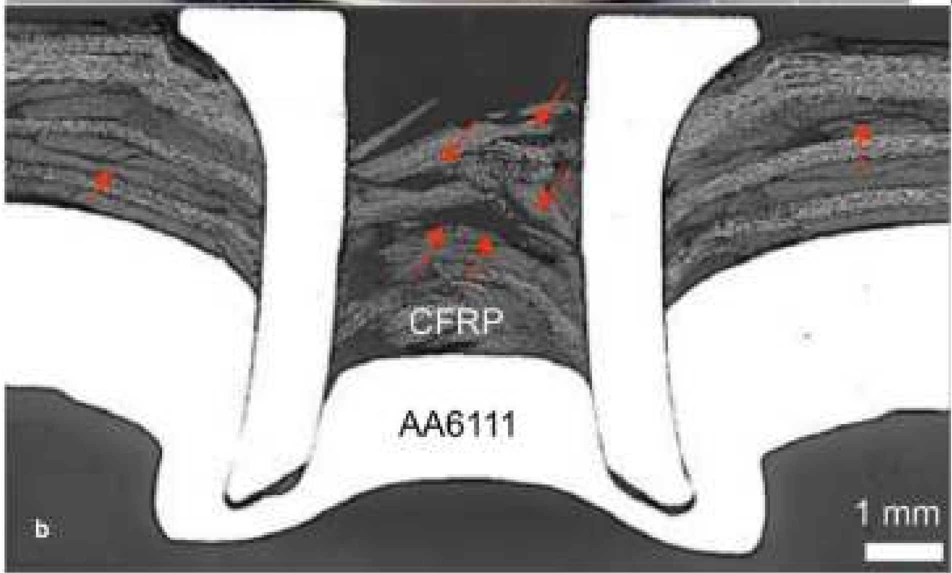
Solid State

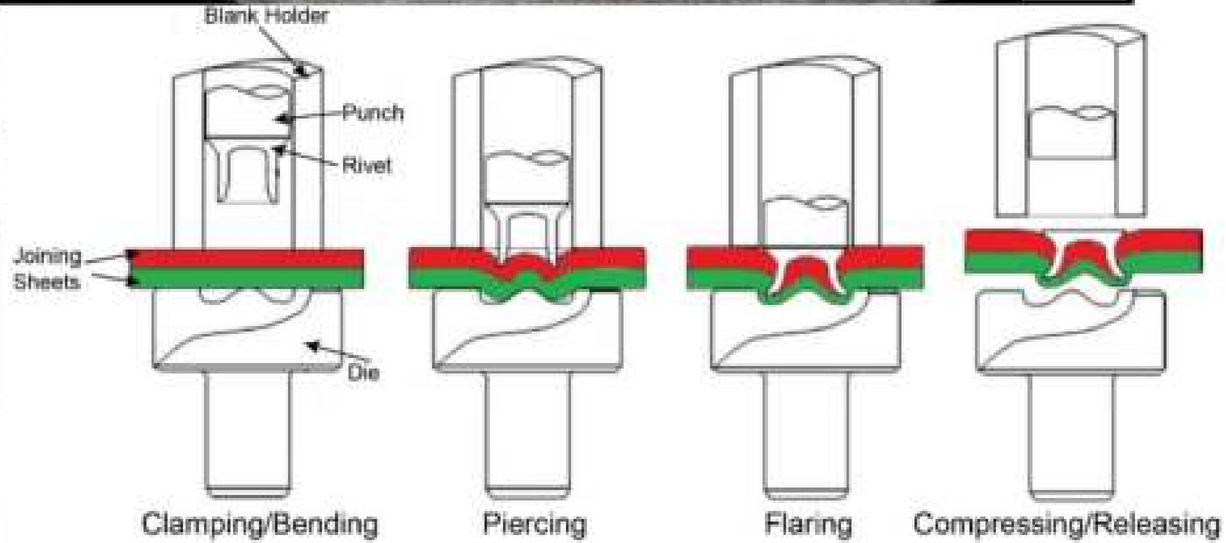
Soldering Electronic Components

More on Soldering

Joints by Self piercing rivets

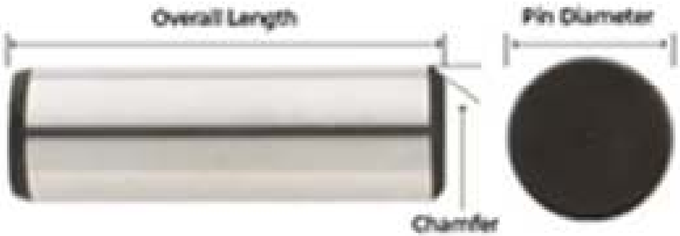






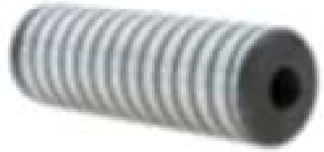
Fasteners

- a hardware device that mechanically joins or affixes two or more objects together.

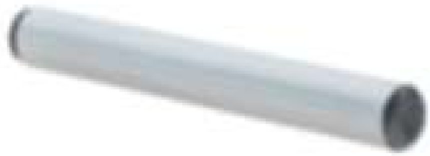


Note: Some Dowel Pins have a Crown Height and Crown Diameter.

Precision dowel pins are effective in interference fit applications. The dowel pin is specially designed based on hole size, tolerance and ductility.



Grooved dowel pins feature grooves along the side for various applications. If the grooves run along only a portion of the pin, the smooth end allows the connected unit to rotate freely. These pins allow for materials to pass through during insertion. They are ideal for blind holes.



Oversized, also known as press-fit or repair dowel pins, are 0.001 inches larger than standard dowel pins. This allows for damaged holes to be opened wider to accommodate the new pin. They are ideal for fastening into out-of-round holes or softer materials.

Pull-Out dowel pins feature an internally threaded hole to allow for removal and reuse with a standard screw or pull-out tool. They are ideal for blind holes. A corresponding wrench fits into the dowel pin hole for easy removal. Pull-out dowels are not referenced by any standard of ASTM or ASME, but by custom, have the same hardness as hardened ground machine dowel pins.

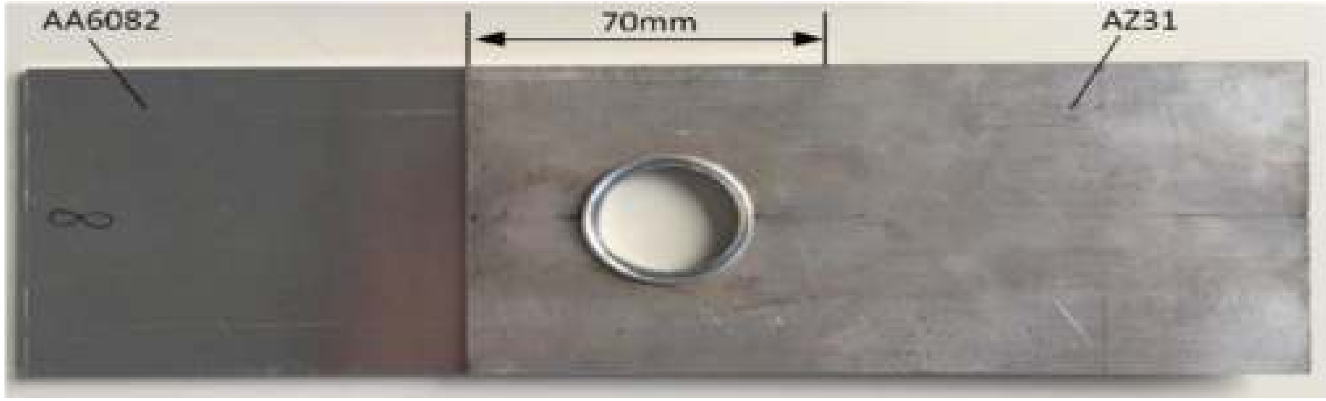
Types of fasteners

Taper Pins have one end slightly larger in diameter than the other, which results in a uniform taper for simple, effective fastening. They create a stop or wedge in machinery applications where frequent assembly/disassembly is required.

Spring Pins, also known as roll pins, are hollow, self-retaining fasteners that secure multiple components and parts together. They are used in light-duty applications in place of solid dowel pins.

Pilot Pins are self-clinching pins that easily embed in sheet metal and other soft materials. They are inserted into mounting holes for alignment applications.

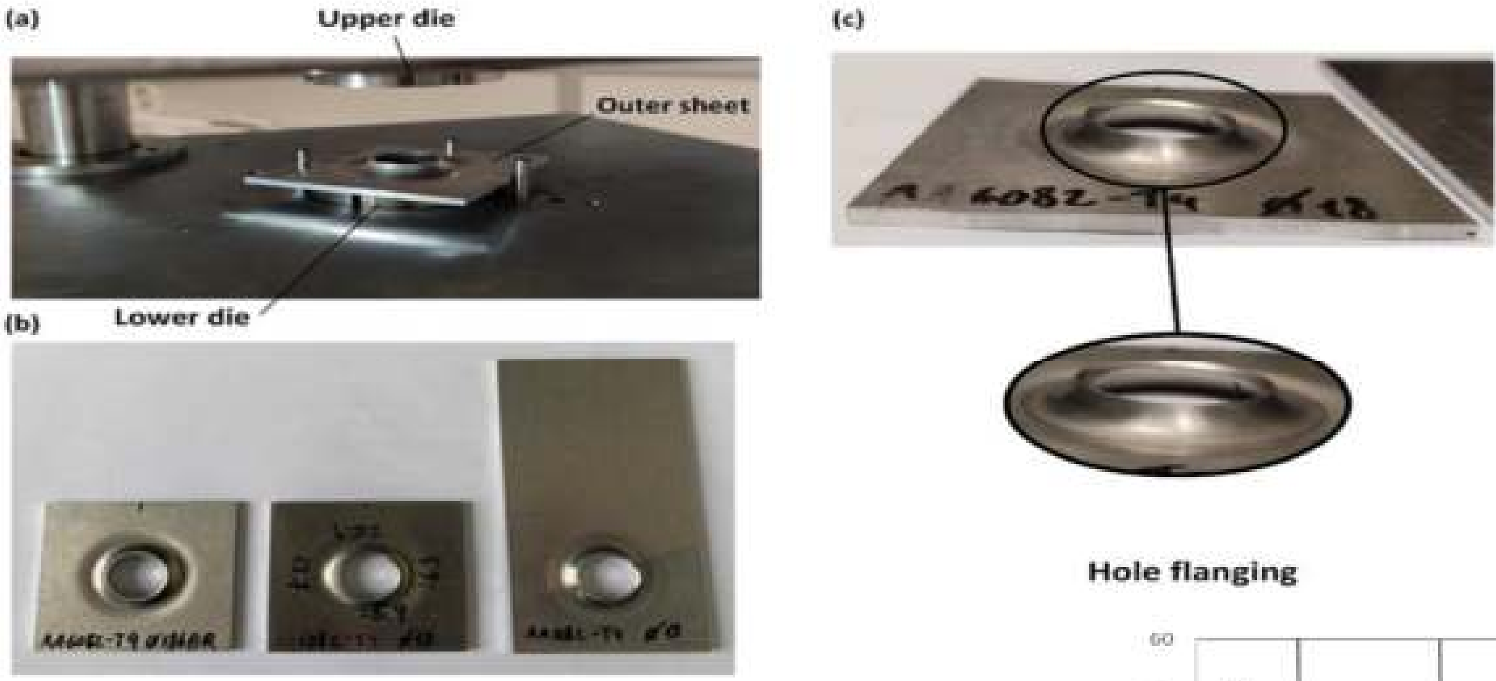
Mechanical Fastening : Hole flanged joint

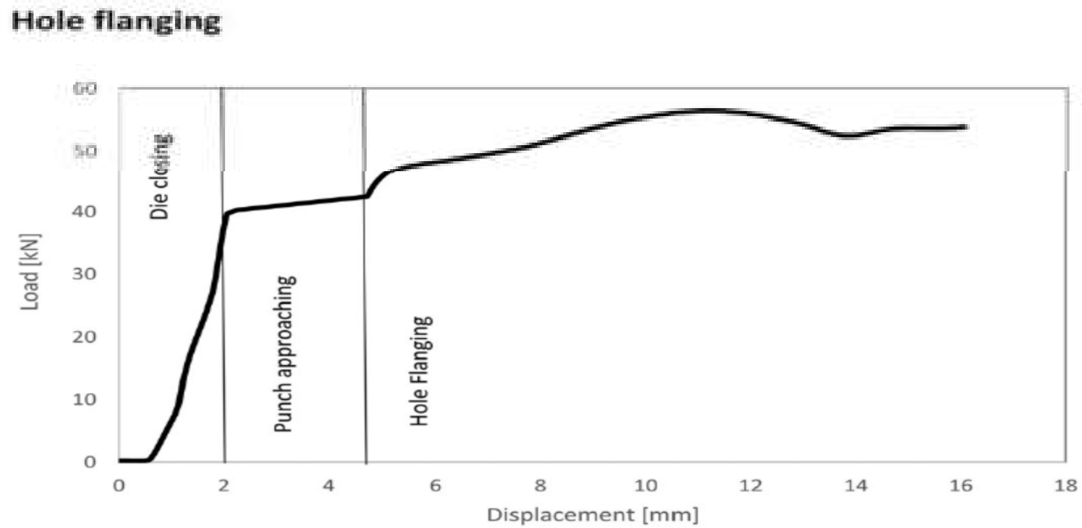


1. Joining of thin materials
2. When at least one material has poor ductility
3. At least one material is highly heat sensitive (may explode on heat application)
4. At least one material has ultra fine grain size (heat rapidly coarsens the grain size)

J. A. C. Pareira, M. M. Kasaei, R. J. C. Karbas, E. A. S. Marques, H. Lim, L. F. M. Da Silva, Joining magnesium and aluminum alloy sheets by a novel hole hemming Process, Thin-Walled Structures 187 (2023) 110758

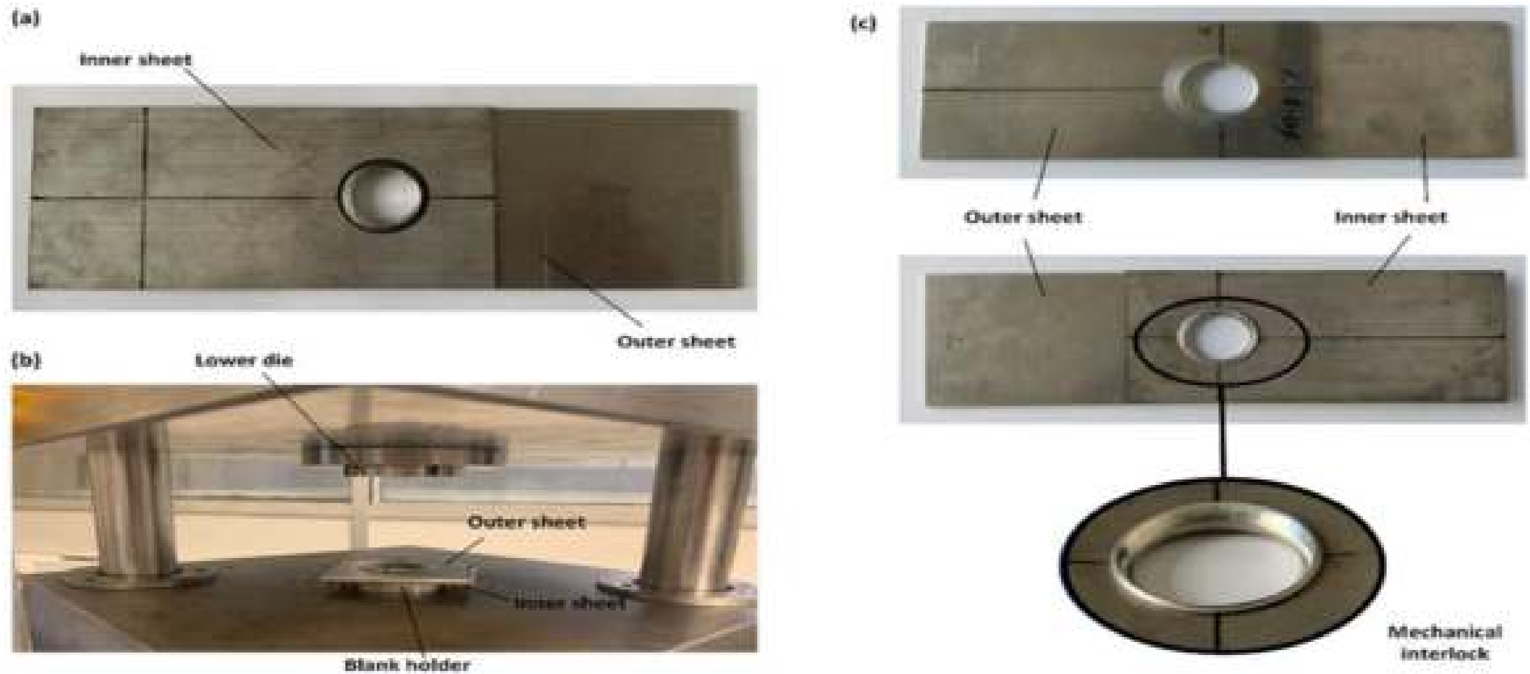
Step 1: Hole Flanging



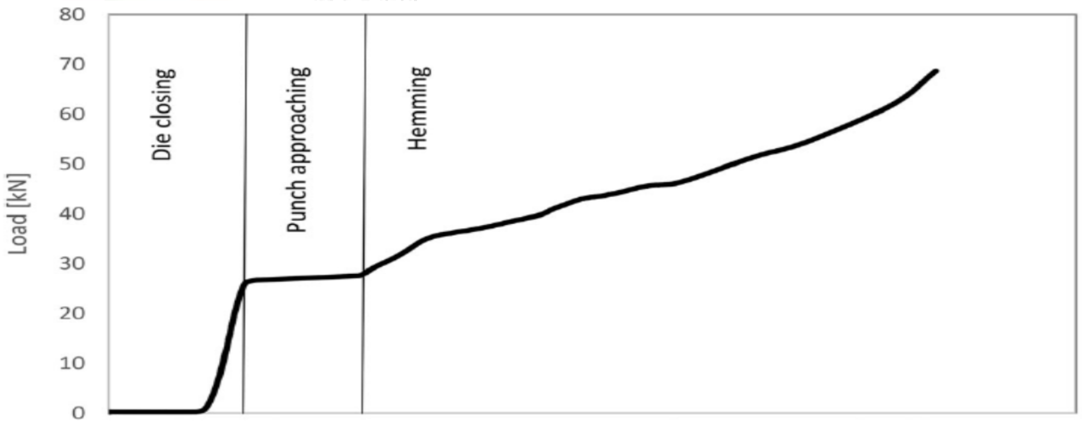


A. G. C. Conceição, M. M.Kasaei, Ricardo J. C. Carbas, E. A. S. Marques, L. F. M. da Silva ,A flexible tool for joining dissimilar materials by the novel hole hemming process, Mechanics Based Design of Structures and Machines, 52(2024)3106-3134

Step 2 : Hemming

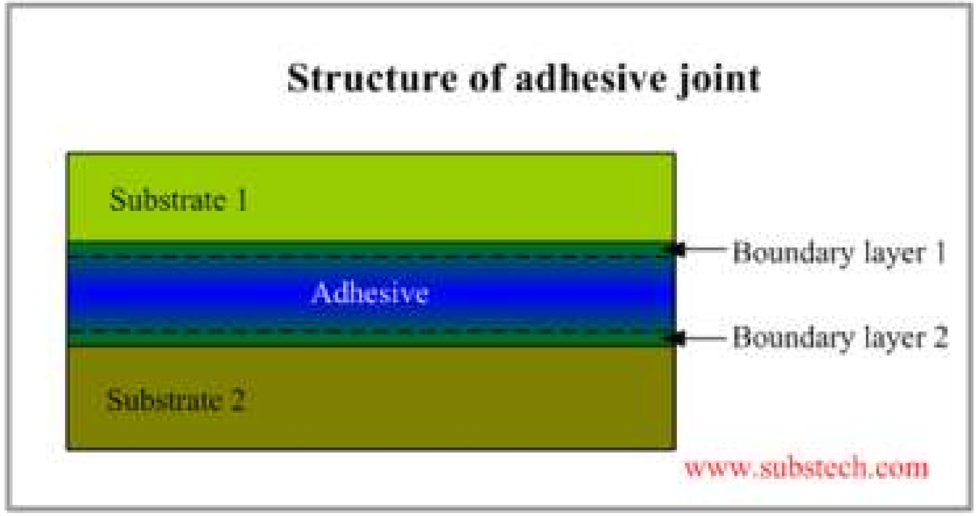


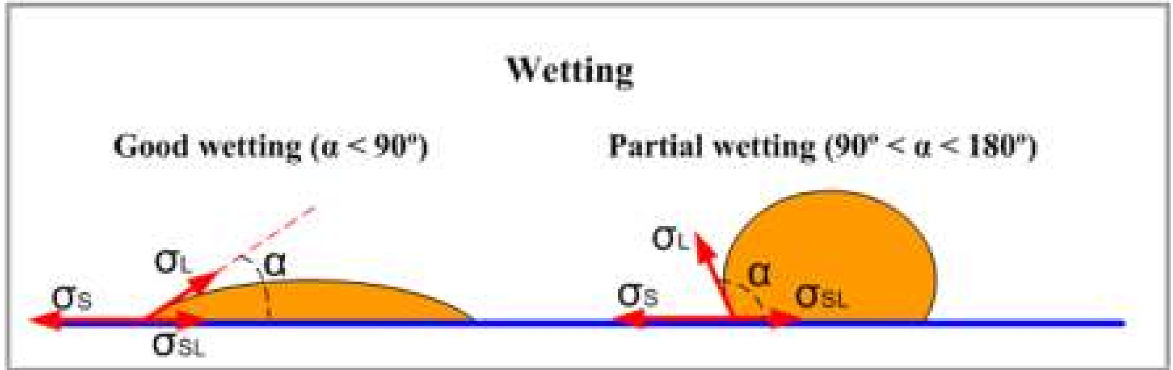
(a) Assembly of the inner sheet on the outer sheet, (b) positioning of the assembly on the blank holder by the pins in the hemming stage and (c) the resulting joint.

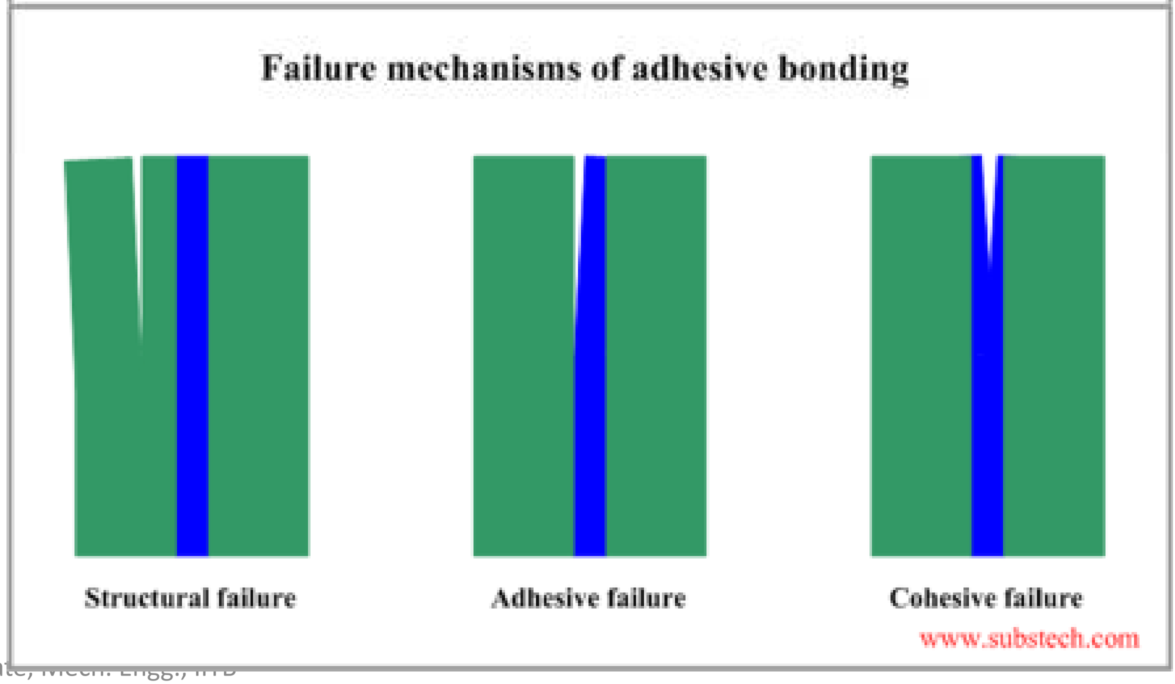


A. G. C. Conceição, M. M.Kasaei, Ricardo J. C. Carbas, E. A. S. Marques, L. F. M. da Silva ,A flexible tool for joining dissimilar materials by the novel hole hemming process, Mechanics Based Design of Structures and Machines, 52(2024)3106-3134

Adhesive Joints (largely for non-metallic materials, dissimilar materials)







Degrease, Abrading, Chemical pretreatment
Physical pretreatment

Characteristics of some adhesives
Epoxy (Single component high strength epoxy adhesive, Two component fast curing epoxy adhesive, Two component toughened epoxy adhesive)
Very good resistance to chemicals (acids, alkalis);
Very good resistance to Solvents and oils; Good resistance to water;
Very good thermal resistance; Strong durable bonding;
Low shrinkage; Ability to fill wide gaps;
Brittleness (poor plasticity);

Summary

- New expectations from a product lead to new materials and new designs
- New materials need new manufacturing processes
- New processes need new equipment
- Alternate manufacturing routes for a given component must be explored.
- Processes like casting, forging, additive manufacturing, laser cutting and a variety of joining methods have been discussed.
- Common manufacturing processes and those used in MS 101 have been discussed.
- Manufacturing as an activity has been introduced

Useful Resources…

- Principles of Modern Manufacturing: Materials, Processes, and Systems, Mikell P. Groover, Wiley India Edition, 2018.
- Manufacturing Engineering and Technology (SI Edition), S. Kalpakjian and S. R. Schmid, Pearson Education; Seventh edition, 2018.
- Fusion 360 Tutorials on additive manufacturing
  - https://help.autodesk.com/view/fusion360/ENU/courses/AP-MFG-ADD-FFF
- How to 3D print using Fusion 360
  - https://www.youtube.com/watch?v=wPScDWi-X4s (practice upto time 3:05)